Understanding Cluster Analysis through Python Libraries

Cluster analysis is a popular technique in data analysis and exploration that finds similarities between different groups of data, based on which datasets are classified or segmented into predefined clusters. It is an unsupervised machine learning algorithm and doesn't need data that has been previously categorised or labelled. Instead, the algorithm identifies patterns and structures within the data on its own. Cluster analysis is widely used in various fields such as marketing, biology, finance, and social sciences for tasks like customer segmentation, anomaly detection, pattern recognition, etc.

Python, with its extensive libraries such as scikit-learn, SciPy, and PyClustering, provides a robust platform for implementing cluster analysis algorithms effortlessly. Its simplicity, versatility, and rich ecosystem make Python well-suited for conducting cluster analysis and interpreting complex datasets. Additionally, Python's readability and ease of use contribute to its popularity in the machine learning community.

The versatility of cluster analysis makes it indispensable for uncovering hidden structures and relationships in data. This helps analysts to derive actionable insights and make data-driven decisions, as shown in Table 1.

K-means clustering

Hierarchical clustering and K-means clustering are two popular techniques used for clustering data points into distinct groups. While K-means clustering divides data into a predefined number of clusters (which is represented by the letter 'K'), hierarchical clustering creates a hierarchical tree-like structure to represent the relationships between the clusters. K-means clustering remains a popular method for cluster analysis, where the datasets are clustered into K clusters in such a way that the sum of the squared distances between the clusters and their assigned cluster mean is minimised. The Python algorithm